

Shirin Shujaa



Software Engineering | NLP & Responsible AI

Summary

Software Engineering graduate (Honours) with a Minor in AI & ML, specializing in AI and automation with growing research experience in NLP and Responsible AI. Currently contributing to a funded research project on Responsible Generative AI for Low-Resource Languages. Proficient across the AI/ML stack and experienced in full-stack development using Python. I am passionate about AI ethics, fairness, and human-AI interaction, with a focus on developing human-centered and socially responsible AI technologies.

Personal Information

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Education

Nov 2022 – Feb 2026 **Bachelor of Software Engineering (Honours), Minor in Artificial Intelligence & Machine Learning** *Royal Melbourne Institute of Technology (RMIT University)* Ho Chi Minh City, Vietnam.
- Academic Excellence Scholarship Recipient (2022–Present).
- Relevant coursework: Data Structures & Algorithms, Computing Theory, Machine Learning, Artificial Intelligence, Software Architecture & Design, Programming Autonomous Robots, Programming IOT

Professional & Research Experience

Jun. 2025 – Present **RMIT University Vietnam** - *Research Assistant, AI & Health Informatics Research Cluster (Project CC_10229: "Toward Responsible Gen AI for Low-Resource Languages")*.
Conducting NLP research on cultural and linguistic bias in AI-generated text. Designing data pipelines for prompt engineering, model sampling, and dataset documentation. Co-developing bias-aware fine-tuning experiments integrating NLP and sociolinguistic methods. Supporting annotation workflows, literature reviews, and paper writing. Collaborating with VLSP partners on open-source datasets and publications.

Sep. 2025 – Jan 2026 **DIGI-TEXX, Ho Chi Minh City, Vietnam** - *AI & NLP Intern*.
Developing NLP and document-understanding systems for enterprise automation. Training Named Entity Recognition (NER) and text classification models for Vietnamese medical data. Building OCR-based data extraction pipelines and integrating NLP models into internal workflow automation tools.

Mar. 2025 – Sep. 2025 **Intel Vietnam** - *Automation Engineer Intern*.
Led end-to-end automation architecture and VBA integration for custom reporting. Implemented pipelines for file ingestion, email scheduling, and SharePoint archiving, reducing processing time from 4 hours to under 15 minutes while maintaining 100% accuracy.

Jan. 2024 – May 2024 **Wareflex, Ho Chi Minh City, Vietnam** *Front-End Developer Intern*.
Developed a responsive landing page using React.js & Tailwind CSS for a logistics platform. Reviewed and optimized 100+ pull requests aligned with headless architecture and integrated backend APIs for real-time data.

Publications

- 1 Detecting Framing Bias in News via Probabilistic Graphical Modeling. **Shirin Shujaa** & Ginel Dorleon. In *Proc. of ACM/IEEE JCDL 2025*. Accepted.
- 2 PrefixGuard LLMs: Hate Speech and Sexism Detection. **Shirin Shujaa**, Ginel Dorleon & Wissam M. Kouadri. In *Proc. of ACM/IEEE JCDL 2025*. Accepted.
- 3 How LLMs Handle Cultural Bias: Reactions to Asian Minority Historical Narratives. **Shirin Shujaa**, Ginel Dorleon & Arthur Tang. In *Proc. of ICADL 2025*. Accepted.
- 4 Uncovering Cultural Biases and Stereotypes in LLMs. Ginel Dorleon & **Shirin Shujaa**. In *Proc. of ICADL 2025*. Accepted.

- 5 ASAP: Constant-Factor Proportional Fairness via Learning-Augmented Online. Ginel Dorleon & **Shirin Shujaa**. In *Proc. of IEEE ICDM 2025 (RDM Workshop)*. Accepted.

AI & ML Projects

- Jun. 2025 **PosePlay** *Gesture-Controlled Game (Computer Vision)*.
Built a gesture-controlled game using OpenCV and MediaPipe for real-time hand tracking and in-game actions.
- May 2025 **Autonomous Robotics Navigation** *RMIT University*.
Built an autonomous navigation system on the mBot2 Neo using a proportional control algorithm, colour-based traffic logic, and real-time obstacle avoidance.
- Mar. 2025 **House Price Prediction** *Kaggle Competition*.
Ranked in the top 2.86% using CatBoost and feature engineering on 81 features.
- Nov. 2024 **Furniture Detection & Classification** *Computer Vision*.
Fine-tuned InceptionV3 for furniture detection and classification, achieving 96.6% accuracy on real-world images.
- Nov. 2024 **Maze Solving Algorithm** *Algorithmic Pathfinding*.
Implemented BFS, DFS, and A* for grid-based pathfinding and cost optimisation.

Engineering & Software Projects

- Jul. 2025 **Content Discography** *Full-Stack Music Archive*.
Developed a full-stack app (React, Node.js, Supabase) to visualise artist timelines and discographies via REST APIs and a responsive UI.
- Sep. 2024 **BuZznet** *Full-Stack Social Platform*.
Built authentication, group management, and admin dashboards with React, Redux, Node.js, MongoDB, and JWT-based access control.
- May 2024 **Mario Game for Bare-Metal OS** *Low-Level C/Assembly Project*.
Implemented obstacle mechanics and enemy AI using GPIO on Raspberry Pi with hardware-level I/O control.

Skills

- Programming** Python, C/C++, JavaScript
- AI & ML** NLP & vision model training, fine-tuning, and evaluation (PyTorch, TensorFlow, Scikit-learn, Hugging Face)
- Data** Preprocessing, analysis, visualisation, and experiment tracking with pandas, NumPy, Matplotlib
- Design** Interface design, data visualisation, and rapid prototyping with Figma, Tailwind CSS, Adobe Illustrator
- Tools** Git, Docker, Linux, Jupyter, VS Code

Awards & Scholarships

- Academic Excellence Scholarship – RMIT University (2022–Present)
- SIGIR Student Travel Grant, ACM/IEEE JCDL 2025
- Selected for SPARK Hub Incubation Program, RMIT University (2025)
- 1st Place – CoverGo AI Hackathon (2025)
- 2nd Place – RMIT × Katalon Testathon (2025)
- 2nd Place – ADC Competition (2025)

Certifications

- ISTQB Certified Tester (Nov 2025)
- Stanford University (Coursera) – Supervised Machine Learning; Generative AI for Everyone
- Microsoft (Coursera) – Fundamentals of UI/UX Design
- UC Irvine (Coursera) – Introduction to IoT & Embedded Systems

Languages

- English Fluent (IELTS 8.0/9.0), TESOL Certified
- Arabic Native
- French Intermediate (B1)

Others

-  ACM SIGIR Member & Funding Recipient (2025)
-  ICADL 2025 Session Chair